

Real Analysis Problem Set

Problem 1

Let $\{a_n\}$ be a sequence of real numbers. Prove that if $\lim_{n \rightarrow \infty} a_n = L$, then $\lim_{n \rightarrow \infty} |a_n| = |L|$.

Proof:

Given that $\lim_{n \rightarrow \infty} a_n = L$, we know that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have $|a_n - L| < \varepsilon$.

We want to show that $\lim_{n \rightarrow \infty} |a_n| = |L|$, which means for every $\varepsilon > 0$, there exists $M \in \mathbb{N}$ such that for all $n \geq M$, we have $||a_n| - |L|| < \varepsilon$.

Using the reverse triangle inequality: $||a_n| - |L|| \leq |a_n - L|$

Since $\lim_{n \rightarrow \infty} a_n = L$, for any $\varepsilon > 0$, there exists N such that $|a_n - L| < \varepsilon$ for all $n \geq N$.

Therefore, for all $n \geq N$:

$$||a_n| - |L|| \leq |a_n - L| < \varepsilon$$

Hence, $\lim_{n \rightarrow \infty} |a_n| = |L|$. $\square$

Problem 2

Let $f: [0,1] \rightarrow \mathbb{R}$ be defined by:

$$f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

Determine whether f is Riemann integrable on $[0,1]$.

Solution:

We will show that f is Riemann integrable and $\int_0^1 f(x) dx = 0$.

Recall that a function is Riemann integrable if and only if the set of discontinuities has measure zero.

Let's analyze the discontinuities of f :

- At any irrational point x_0 : For any $\delta > 0$, we can find a rational number r in $(x_0 - \delta, x_0 + \delta)$. Since $f(r) = r^2 > 0$ (for $r \neq 0$) and $f(x_0) = 0$, we have $|f(r) - f(x_0)| = r^2$. Since we can make r arbitrarily close to x_0 , f is discontinuous at every irrational point.
- At any rational point x_0 : Similarly, we can find irrational numbers arbitrarily close, and $f(\text{irrational}) = 0$ while $f(x_0) = x_0^2$, so f is also discontinuous at every rational point.

However, the set of discontinuities is $[0,1]$, which does NOT have measure zero in the usual sense.

But wait—let's use a different approach. We'll show that the upper and lower Riemann sums converge to the same

value.

For any partition P of $[0,1]$:

- Lower sum $L(f,P) = 0$ (since every subinterval contains irrationals where $f = 0$)
- Upper sum $U(f,P) = \sum M_i \Delta x_i$, where M_i is the sup of f on the i -th subinterval

Since rationals are dense, $M_i = (x_i)^2$ where x_i is the right endpoint. As the partition mesh $\rightarrow 0$, the upper sum approaches $\int_0^1 x^2 dx = 1/3$.

Since the lower sum is always 0 and the upper sum approaches $1/3$, they don't converge to the same value. Therefore, f is NOT Riemann integrable.

Actually, let me reconsider. The function is discontinuous everywhere, and the set of discontinuities is $[0,1]$ which has positive measure. By Lebesgue's criterion, f is NOT Riemann integrable.

Problem 3

Prove that if f is continuous on $[a,b]$ and differentiable on (a,b) , and $f(a) = f(b)$, then there exists $c \in (a,b)$ such that $f'(c) = 0$.

Proof:

This is a statement of Rolle's Theorem. Here is the proof:

Since f is continuous on the closed interval $[a,b]$, by the Extreme Value Theorem, f attains both its maximum and minimum on $[a,b]$.

Case 1: If the maximum or minimum occurs at an interior point $c \in (a,b)$, then by Fermat's Theorem, $f'(c) = 0$ (since c is a local extremum in an open interval).

Case 2: If both the maximum and minimum occur at the endpoints a and b , then since $f(a) = f(b)$, we have $f(x) = f(a) = f(b)$ for all $x \in [a,b]$. This means f is constant, so $f'(x) = 0$ for all $x \in (a,b)$, and we can choose any $c \in (a,b)$.

In either case, there exists $c \in (a,b)$ such that $f'(c) = 0$. $\square$

Problem 4

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be uniformly continuous. Prove that if $\{x_n\}$ is a Cauchy sequence, then $\{f(x_n)\}$ is also a Cauchy sequence.

Proof:

Let $\{x_n\}$ be a Cauchy sequence. We want to show that $\{f(x_n)\}$ is a Cauchy sequence, i.e., for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, we have $|f(x_m) - f(x_n)| < \varepsilon$.

Since f is uniformly continuous, for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in \mathbb{R}$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.

Since $\{x_n\}$ is Cauchy, for the δ found above, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, we have $|x_m - x_n| < \delta$.

Therefore, for all $m, n \geq N$:

$$|x_m - x_n| < \delta \implies |f(x_m) - f(x_n)| < \varepsilon$$

Hence, $\{f(x_n)\}$ is a Cauchy sequence. $\square$

